VISION-ASSISTED ROBOTIC SOLDERING

KLE Technological University | Creating Value Leveraging Knowledge

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PROBLEM STATEMENT

Manual PCB soldering can lead to **defects, inconsistencies, and human error.** This project proposes a Vision-Assisted Robotic Soldering System that combines computer vision and robotic automation for **accurate PCB inspection** and **automated soldering.**

AUTOMATION | VISION | ACCURACY | QUALITY

Objectives

- **Automate** PCB soldering operations using robotic systems
- **Detect** and **verify** PCB components using **computer vision**
- Improve soldering **accuracy** and **consistency**
- **Reduce** manual intervention and production defects

AUTOMATION | VISION | ACCURACY | QUALITY

REQUIREMENTS (Selected)

FUNCTIONAL

- Capture PCB Image
- PCB Detection
- XYZ Motion Control

NON-FUNCTIONAL

- Real-time Processing
- Stable Motion
- Consistent Performance

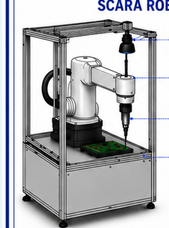
SPECIFICATIONS

Storage Capacity	≥128 GB
Camera Resolution	12 MP
Position Accuracy	±1 mm
Workspace	2 × 2 × 2 ft
Motion Axes	X-Y-Z
Load Capacity	Up to 3 kg
Operating Temperature	250–350°C
Processing Speed	15–25 FPS
Continuous Operation	4–6 Hours

CONCEPTUAL DESIGNS & CONCEPT SELECTION

Vision-Assisted Robotic Soldering Process Optimization For Electronic and Light Fabrication Applications

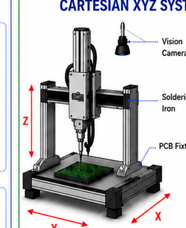
DESIGN 1 – SCARA ROBOT



FEATURES: High-speed motion; Vision-assisted inspection; Enclosed safety cabinet; Industrial production focus

APPLICATION: High-throughput production lines

DESIGN 2 – CARTESIAN XYZ SYSTEM (SELECTED DESIGN)



FEATURES: Cartesian XYZ structure; High positional accuracy; Simple rigid frame; Easy maintenance; Scalable workspace

APPLICATION: General-purpose PCB soldering with high accuracy & reliability

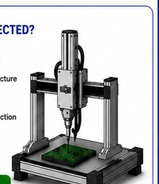
PUGH CHART – CONCEPT SELECTION

CRITERIA	WEIGHT (1–5)	DESIGN 1 SCARA	DESIGN 2 CARTESIAN XYZ
Precision & Accuracy	5	+	+
Flexibility	4	0	+
Cost Effectiveness	4	0	+
Ease of Maintenance	4	–	+
Workspace / Scalability	3	0	+
Vision Integration	3	0	+
Implementation Complexity	4	–	0
TOTAL SCORE (Weight × Score)		+1	+7

+ : Better 0 : Same – : Worse

WHY DESIGN 2 IS SELECTED?

- Highest overall score in Pugh chart (+7)
- High precision with simple and rigid structure
- Lower cost and easy to maintain
- Scalable and suitable for industrial production
- Best balance between performance, cost and reliability



FINALIZED DESIGN: DESIGN 2 – CARTESIAN XYZ SYSTEM

Selecting DESIGN 2 ensures accurate, reliable and scalable robotic soldering for electronic toy manufacturing at Aequs Koppal.

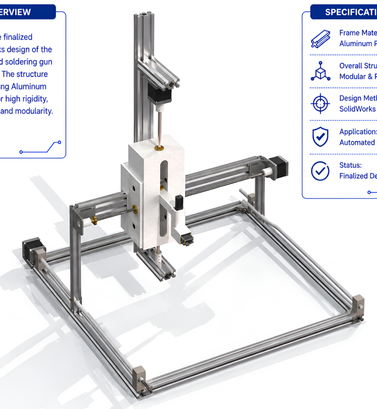
FINALIZED SOLIDWORKS DESIGN

OVERVIEW

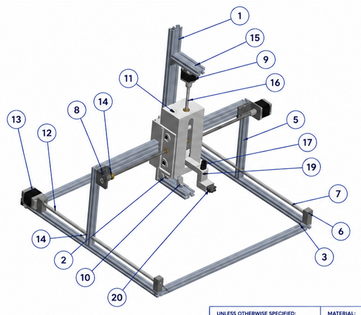
This is the finalized SolidWorks design of the automated soldering gun assembly. The structure is built using Aluminum profiles for high rigidity, precision, and modularity.

SPECIFICATIONS

- Frame Material: Aluminum Profile
- Overall Structure: Modular & Rigid
- Design Method: SolidWorks
- Application: Automated Soldering
- Status: Finalized Design



BILL OF MATERIAL



ITEM NO.	PART NUMBER	QTY.
1	ALUMINIUM_PROFILE_20m x 20mm x 500mm	8
2	LBS86-2020_b	11
3	IS 2269 - M4 x 8-S	16
4	IS 2015 - 5-S	8
5	ALUMINIUM_PROFILE_20m x 20mm x 500mm - Copy	2
6	ROD_HOLDER BLOCK	5
7	smooth_rod	2
8	clamp stepmotor nema 17	2
9	Nema 17	2
10	Flexible Shaft Coupler 5x8mm.stop	4
11	Aluminium_Profile	1
12	Lead Screw D8 mm, 2mm Pitch (2019)	2
13	Nema 17 - stepper motor	1
14	LeadScrew Nut 8mm x 2mmPitch	4
15	Z-Axis_Carriage	4
16	Trapezoidal lead screw tr8x8-4	1
17	Soldering_Gun	1
18	Soldering gun holder/Assem1	1
19	Soldering_Mount	1
20	ESP32-CAWI-ESP32CA-WI-MB.step	1

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS; SURFACE FINISH: –; TOLERANCES: LINEAR: ±0.5 mm; ANGULAR: ±1°

MATERIAL: –

DEBURR AND BREAK SHARP EDGES

	NAME	SIGNATURE	DATE
DRAWN	Shahista		
CHK'D	–	–	–
APPV'D	–	–	–
MFG	–	–	–
Q.A.	–	–	–

WEIGHT: – SCALE: 1:5 SHEET 1 OF 1

DO NOT SCALE DRAWING REVISION: –

TITLE: Assembly

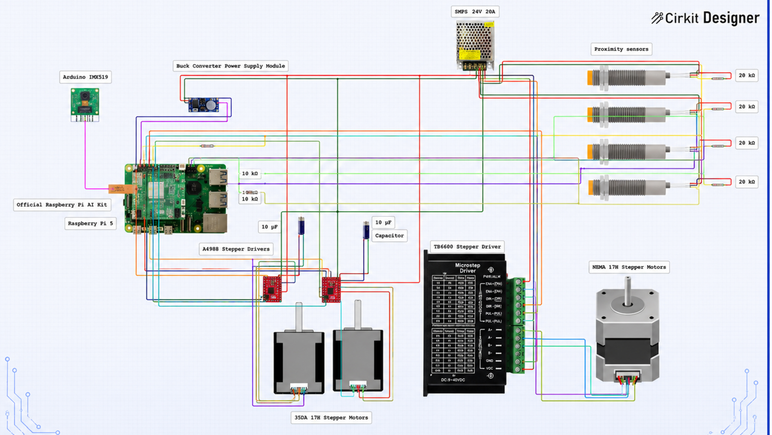
DWG NO.: Team_03 A3

PROCESS FLOW

VISION-ASSISTED ROBOTIC SOLDERING PIPELINE

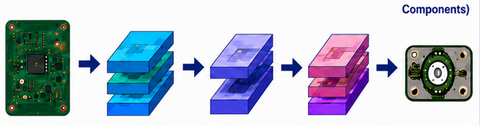
PCB Image Input
↓
Preprocessing
↓
YOLOv8 Detection
↓
Front PCB Component Detection | Back PCB Hole Detection
↓
Coordinate Extraction
↓
A* Path Planning
↓
3-DOF Cartesian Robotic Soldering

CIRCUIT DESIGN



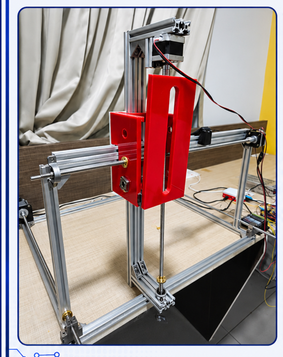
SOFTWARE IMPLEMENTATION

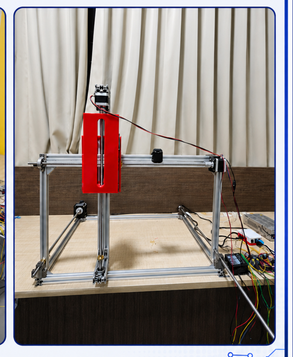
YOLO ARCHITECTURE



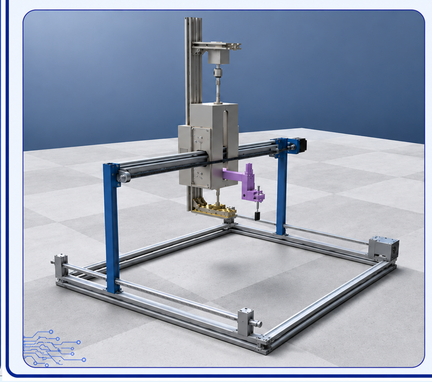
DETECTION RESULTS

HARDWARE





SIMULATION



SOFTWARE : CoppeliaSim

- 3D Robot Modeling & Simulation
- Accurate Kinematics & Motion Analysis
- Collision Detection & Path Validation
- Realistic Environment & Performance Analysis

PORTFOLIO

(Scan to Deatails)